Week 10: Credibility Theory

F79MA Statistical Models A

Heriot-Watt University

Introduction

Motivating Example

Suppose that the local council of the small town of Tinyville wants to insure its five public parks against flood, fire and malicious damage. Upon extensive data collection, it was found that the average annual claim amount (for all five parks combined) over the past 2 years was \$10,000. In addition to this information, the insurer has data from many other small local councils from all over the country, which reveals that the average annual claim amount per park was \$3,000.

Question: How much pure premium should the insurer charge for the next year?

Solution: It is worth noting that we can adopt one of the following two extreme strategies in setting the pure premium:

- Strategy 1: Set the premium as \$10,000, i.e. we rely entirely on the information provided by the local council concerned.

  Note: *The information/data are directly relevant, but may not be sufficient.*

- Strategy 2: Set the premium as \$3,000 $\times$ 5 = \$15,000, i.e. we rely entirely on the data gathered from different local councils around the country.

  Note: *There is an abundance of information/data, but they may not be relevant (or they may be too 'noisy').*

It turns out that we can adopt a third strategy, that is to set the premium as a weighted average of the two extremes above. This is known as the **credibility** approach. Mathematically, we can calculate the pure premium (usually called the **Credibility Premium**) as

$$10,000 \times Z + 15,000 \times (1 - Z),$$

where $0 \leq Z \leq 1$ is known as the credibility factor.

Intuitively, the credibility approach aims to take into account not only the individual losses, but also the collective losses which may reflect certain *prior information* or *knowledge* about the losses.

Credibility Theory

Now let's try to put all these more formally in the actuarial context. The aim of Credibility Theory is to provide an estimate for the expected aggregate claims (or possibly, the expected number of claims) in the coming year from a risk. By risk, we mean a single (insurance) policy or a group of policies. These policies are usually short-term policies (typically 1 year).

The following information is available to us:

- Data from the risk itself, with mean $\bar{X}$.
- Data from a larger group of similar, but not necessarily identical, risks (known as "collateral data") with mean μ.

The credibility premium or credibility estimate for this risk is:

$$Z\bar{X} + (1 - Z)\mu, \tag{1}$$

where $0 \leq Z \leq 1$ is the credibility factor.

In the actuarial context, the credibility factor Z measures our trust (or belief) in the credibility of the data from the risk itself (i.e., $\bar{X}$). It will be shown later that, the more directly relevant data we have, the higher the value of the credibility factor (and thus the weighting given to $\bar{X}$).

It is worth noting that the calculation of the credibility premium has connections with *Bayesian* estimates. Recall from the Week 8 Reading Material that the posterior expectation is the optimal estimator for a Bayesian decision problem based on the minimisation of the mean square error loss. In fact, for certain parametric models, the Bayesian posterior expectation can be expressed in the form (1). However, for most other Bayesian models, the posterior expectation is generally not a credibility estimate. In those cases, we can see (1) as a computationally efficient approximation of the minimum mean square error estimator.

Motivating Example (continued)

(a) Suppose that $Z = 0.6$, what would the credibility premium be?

$$0.6 \times 10000 + 0.4 \times 15000 = 12000$$

(b) Suppose the data we have on Tinyville's claims were for 10 years instead of 2 years. This would imply that Tinyville's data is more relevant than the collateral data, and therefore Z would increase to perhaps 0.75. What would the credibility premium be?

$$0.75 \times 10000 + 0.25 \times 15000 = 11250$$

(c) Suppose the data we have on Tinyville's claims were for 2 years, but the collateral data obtained were from parks of very similar types to those in Tinyville. This would imply that Tinyville's data are less relevant than the collateral data, and therefore Z would decrease to perhaps 0.4. What would the credibility premium be?

$$0.4 \times 10000 + 0.6 \times 15000 = 13000$$

Now that we have seen a few examples for different values of Z. However, the problem remains of how to derive a *well-justified* value for the credibility factor Z. There are generally two approaches:

- Bayesian approach to credibility, and
- Empirical Bayes Credibility Theory (EBCT).

We will visit these in the subsequent sections.

Bayesian Approach to Credibility

This approach involves similar steps and ideas as in Bayesian estimation which we have seen previously. We need the following components:

1. **Prior:** It is derived from the information about the underlying collateral data.
2. **Likelihood:** It is derived from a parametric model and historical data from the risk itself.
3. **Posterior:** It is proportional to Likelihood $\times$ Prior.

We will go through the following two Bayesian credibility models:

- the Poisson/gamma model (for claim frequencies), and
- the normal/normal model (for claim amounts).

The Poisson/gamma model

Let the random variable X denote the number of claims in the coming year from a certain risk. Suppose that the number of claims each year follows a Poisson distribution with parameter λ, i.e., $X|\lambda \sim \text{Poi}(\lambda)$, and suppose that λ is unknown. It makes sense to take the distribution $\text{Gamma}(\alpha, \beta)$ as the (conjugate) prior for λ, where α and β are known constants.

We assume that data are available for the past n years from this risk, which means that we have n observations of X, i.e., $\underline{\text{x}} = (x_1, x_2, \cdots, x_n)$. It can be shown that the posterior distribution of λ given $\underline{\text{x}}$ is

$$(\lambda|\underline{\text{x}}) \sim \text{Gamma}\left(\alpha + \sum_{i=1}^{n} x_i, \beta + n\right),$$

and that the Bayes estimate under mean square error loss (i.e. the posterior expectation) is

$$\mathbf{E}(\lambda|\underline{\mathrm{x}}) = \bar{x}Z + \mu(1-Z),$$

where

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \ , \quad Z = \frac{n}{\beta+n} \ , \quad \text{and} \quad \mu = \frac{\alpha}{\beta}.$$

Note that the Bayes estimate above resembles the form of a credibility estimate/premium.

Remarks:

1. Z reflects the relative reliability between the two extreme estimates for λ (one is completely reliant on the data, the other one is completely reliant on our prior information).

2. If no data are available from the risk (i.e., $n = 0$), then we have $Z = 0$, and the credibility estimate is just equal to the prior mean μ.

3. If the only data we use are from the risk itself, then the credibility estimate would be equal to $\bar{x}$, which is precisely the MLE for λ. Note that the estimate itself is a linear combination of the observed data values.

4. Z increases (to one) as the sample size n increases. The more data we obtained from the risk itself, the higher the credibility factor would be.

5. Z decreases (to zero) as β increases.

Example 1 Suppose that the annual number of claims X resulting from a particular risk in a small town follows a Poi(λ) distribution. Collateral data suggest a prior distribution of Gamma(100,1). The table below shows the annual number of claims over the past 5 years. Calculate the credibility estimates for each year.

Year	1	2	3	4	5
No. of Claims	144	160	152	156	158

Solution: We work out the credibility estimate for each year in the following table:

Year (n)	Credibility factor, $Z = \frac{n}{\beta+n}$	Sample average, $\bar{x}$	Credibility Estimate, $\bar{x}Z + (1-Z)\frac{\alpha}{\beta}$
0	$\frac{0}{1+0} = 0$	–	$0 + (1-0) \times 100 = 100$
1	$\frac{1}{1+1} = \frac{1}{2}$	144	$144 \times \frac{1}{2} + \left(1 - \frac{1}{2}\right) \times 100 = 122.00$
2	$\frac{2}{1+2} = \frac{2}{3}$	$\frac{144+160}{2} = 152$	$152 \times \frac{2}{3} + \left(1 - \frac{2}{3}\right) \times 100 = 134.67$
3	$\frac{3}{1+3} = \frac{3}{4}$	$\frac{144+160+152}{3} = 152$	$152 \times \frac{3}{4} + \left(1 - \frac{3}{4}\right) \times 100 = 139.00$
4	$\frac{4}{1+4} = \frac{4}{5}$	$\frac{144+160+152+156}{3} = 153$	$153 \times \frac{4}{5} + \left(1 - \frac{4}{5}\right) \times 100 = 142.40$
5	$\frac{5}{1+5} = \frac{5}{6}$	$\frac{144+160+152+156+158}{5} = 154$	$154 \times \frac{5}{6} + \left(1 - \frac{5}{6}\right) \times 100 = 145.00$

The normal/normal model

Let the random variable X denote the monetary amount of the aggregate claim in the coming year from a certain risk. Suppose that the aggregate claim amount follows a normal distribution with parameter θ, i.e., $X|\theta \sim N(\theta, \sigma_1^2)$, and suppose that θ is fixed but unknown. We can take the prior distribution $\theta \sim N(\mu, \sigma_2^2)$. Here, μ, σ_1^2, and σ_2^2 are known. Note that σ_1^2 and σ_2^2 have different interpretations:

- σ_1^2 is known as the **within risk variance**. It can be interpreted as the variance of the claim amounts across different years of the same risk, i.e., how much the company's data vary over the years of data used. A large σ_1^2 would suggest that the individual risk's data vary greatly from year to year, making the predictive inference less reliable.
- σ_2^2 is known as the **between risk variance**. It can be interpreted as the variance between the different risks, i.e., how much each policy data vary from other policy data in the same portfolio. A large σ_2^2 would suggest that the collateral data are too volatile and may contain too much noise.

We assume data are available for the past n years from this risk, which means we have n observations of X, i.e., $\underline{\mathrm{x}} = (x_1, x_2, \cdots, x_n)$. It can be shown that the posterior distribution of θ given $\underline{\mathrm{x}}$ is

$$(\theta|\underline{\mathrm{x}}) \sim N\left(\frac{\frac{1}{\sigma_2^2} \cdot \mu + \frac{1}{\sigma_1^2/n} \cdot \bar{x}}{\frac{1}{\sigma_2^2} + \frac{1}{\sigma_1^2/n}}, \left[\frac{1}{\sigma_2^2} + \frac{1}{\sigma_1^2/n}\right]^{-1}\right),$$

and that the Bayes estimate under mean square error loss, i.e. the posterior expectation, is

$$\mathbf{E}(\theta|\underline{\mathrm{x}}) = \bar{x}Z + \mu(1-Z),$$

where

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \quad \text{and} \quad Z = \frac{n}{n + \frac{\sigma_1^2}{\sigma_2^2}} = \frac{n}{n + \frac{\text{within risk variance}}{\text{between risk variance}}}.$$

Note again that the Bayes estimate above resembles the form of a credibility estimate/premium.

Examples: One can think of X as buildings insurance for a specific type of property and location in Scotland for which we have some historic record $\underline{x}$, and μ could be expert knowledge derived from regional information. Similarly, X could be related to a business travel insurance policy for a specific company for which we have some limited data $\underline{x}$ available, and μ could summarise overall claims experience.

Remarks:
The majority of the inferences from the normal/normal model coincide with those from the Poisson/gamma model. It should be noted that the credibility factor Z increases as σ_2^2 increases, and decreases as σ_1^2 increases. This inference is consistent with our interpretations of σ_1^2 and σ_2^2 earlier. When σ_1^2 is large, it is natural to avoid assigning a large weight to the data obtained directly from the risk itself. When σ_2^2 is large, we would avoid assigning a large weight to the collateral data instead, thus leading to an increased weight assigned to the data from the risk.

Example 2 An insurer believes that individual claim amounts follow a normal distribution with an unknown mean θ and a standard deviation of \$150. Information from the industry suggests that θ should be assumed to follow a normal distribution with mean \$100 and standard deviation \$75. Over the previous year, the insurer collected data from 20 claims that were directly related to the risk, and the claim amounts totalled up to \$2050.

(a) Calculate the credibility factor placed on the sample mean.

(b) Calculate the credibility premium the insurer should charge for the following year.

Solution: We know that $(X|\theta) \sim N(\theta, \sigma_1^2 = 150^2)$ where $\theta \sim N(\mu = 100, \sigma_2^2 = 75^2)$

(a) We calculate the credibility factor as

$$Z = \frac{n}{n + \frac{\sigma_1^2}{\sigma_2^2}} = \frac{20}{20 + \left(\frac{150}{75}\right)^2} = \frac{5}{6}$$

(b) We first calculate the sample mean as

$$\bar{x} = \frac{2050}{20} = 102.5$$

Thus, the credibility premium is calculated as

$$\mathbf{E}(\theta|\underline{x}) = Z\bar{x} + (1 - Z)\mu = \frac{5}{6} \times 102.5 + \frac{1}{6} \times 100 = 102.08$$

Empirical Bayes Credibility Theory Model 1

The discerning reader may have realised that all the previous Bayesian credibility models require specifying the likelihood function and the prior distribution. These distributions usually take a parametric form, and their parameters are set by experts who are knowledgeable about the nature of a claim. When this parametric information is not available, it is possible to proceed with a data-driven Bayesian approach, known as the Empirical Bayes Credibility Theory (EBCT) model. Note that the EBCT is sometimes also called nonparametric estimation of Bühlmann models.

Let us first define the Bühlmann model.

Bühlmann model

Assume that there are N risks in a portfolio of policies. For each risk $i = 1, 2, \cdots, N$ there is a risk parameter θ_i. Suppose that the claim amounts observed for risk i over n time periods are $\underline{\mathrm{X}}_i = (X_{i1}, \cdots, X_{in})$. For simplicity, we assume that all claims are observed for n time periods but this is not necessary.

Risk	Period (j)			
number (i)	1	2	$\cdots$	n
1	X_{11}	X_{12}	$\cdots$	X_{1n}
2	X_{21}	X_{22}	$\cdots$	X_{2n}
$\vdots$	$\vdots$		$\ddots$	$\vdots$
N	X_{N1}	X_{N2}	$\cdots$	X_{Nn}

From the Bayesian perspective, the model makes the following assumptions:

- The set of risk parameters $(\theta_1, \ldots, \theta_N)$ is an i.i.d. sample from some unknown distribution $\pi(\theta)$.
- The set of claim amounts $(X_{i1}, \cdots, X_{in})$ is an i.i.d sample conditionally on θ_i, i.e., i.i.d sample from $\pi(x|\theta_i)$.
- The data-risk pairs $\{\underline{\mathrm{x}}_i, \theta_i\}_{i=1}^N$ are i.i.d. samples from some unknown joint distribution $\pi(\underline{\mathrm{x}}, \theta)$, where $\pi(\underline{\mathrm{x}}, \theta) = \pi(\theta) \prod_{j=1}^n \pi(x_j|\theta)$.

Before we look at the Bühlmann credibility formula, let us define the following quantities:

(i) The conditional expectation of the claim amounts for risk i

$$m(\theta_i) = \mathbf{E}(X|\theta_i) = \int x\pi(x|\theta_i)\ \mathrm{d}x$$

(ii) The conditional variance of the claim amounts for risk i

$$s^2(\theta_i) = \mathbf{Var}(X|\theta_i) = \int [x - m(\theta_i)]^2 \pi(x|\theta_i)\ \mathrm{d}x$$

Note: Both $m(\theta_i)$ and $s^2(\theta_i)$ are random variables that depend on the risk parameter θ_i.

Bühlmann credibility formula

The credibility estimate for risk i takes the form $Z\bar{X}_i + (1-Z)\mu$, where $\bar{X}_i = \frac{1}{n}\sum_{j=1}^{n} X_{ij}$, and the credibility factor is calculated as

$$Z = \frac{n}{n + \frac{\mathbf{E}[s^2(\theta_i)]}{\mathbf{Var}[m(\theta_i)]}}.$$

In the formula above, there are three main structural parameters:

(i) The global expected risk μ of the entire portfolio, which is defined as the marginal expectation of X

$$\mu \triangleq \mathbf{E}(X) = \mathbf{E}[m(\theta_i)] = \iint x\pi(x|\theta)\pi(\theta)\ \mathrm{d}x\mathrm{d}\theta.$$

(ii) The expectation with respect to θ of the conditional variance $s^2(\theta_i)$

$$\mathbf{E}[s^2(\theta_i)] \triangleq \int s^2(\theta)\pi(\theta)\ \mathrm{d}\theta.$$

(iii) The variance with respect to θ of the conditional expectation $m(\theta_i)$

$$\mathbf{Var}[m(\theta_i)] \triangleq \int [m(\theta) - \mu]^2 \pi(\theta)\ \mathrm{d}\theta.$$

Note that the credibility factor Z in the Bühlmann credibility formula has a similar interpretation to Z in the normal/normal model discussed earlier. In particular, $\mathbf{E}[s^2(\theta_i)]$ can be thought of as the within risk variance, whereas $\mathbf{Var}[m(\theta_i)]$ can be thought of as the between risk variance.

When a parametric model is not available, we can estimate the aforementioned quantities using suitable estimators. In particular, we will select the sample mean and sample variance since they are unbiased estimators for the theoretical mean $\mathbf{E}(\cdot)$ and theoretical variance $\mathbf{Var}(\cdot)$, respectively. It is natural to estimate $m(\theta_i)$ and $s^2(\theta_i)$ as follows:

$$\widehat{m(\theta_i)} = \bar{X}_i = \frac{1}{n}\sum_{j=1}^{n} X_{ij},$$

$$\widehat{s^2(\theta_i)} = S_i^2 = \frac{1}{n-1}\sum_{j=1}^{n} \left(X_{ij} - \bar{X}_i\right)^2.$$

In tabular form, the sample mean $\bar{X}_i$ and sample variance S_i^2 for the risk i are calculated by aggregating the claim amounts in each row.

Risk number (i)	Period (j) 1	2	$\cdots$	n	$\bar{X}_i$ $= \frac{1}{n}\sum_{j=1}^{n} X_{ij}$	S_i^2 $= \frac{1}{n-1}\sum_{j=1}^{n}(X_{ij}-\bar{X}_i)^2$
1	X_{11}	X_{12}	$\cdots$	X_{1n}	$\bar{X}_1$	S_1^2
2	X_{21}	X_{22}	$\cdots$	X_{2n}	$\bar{X}_2$	S_2^2
$\vdots$	$\vdots$		$\ddots$	$\vdots$	$\vdots$	$\vdots$
N	X_{N1}	X_{N2}	$\cdots$	X_{Nn}	$\bar{X}_N$	S_N^2

We would also like to estimate the structural parameters $\mathbf{E}[m(\theta_i)]$, $\mathbf{E}[s^2(\theta_i)]$, and $\mathbf{Var}[m(\theta_i)]$. Let us define the grand mean $\bar{X} = \frac{1}{N}\sum_{i=1}^{N}\bar{X}_i$. Then, we can estimate the quantities as:

$$\widehat{\mathbf{E}[m(\theta_i)]} = \bar{X} = \frac{1}{N}\sum_{i=1}^{N}\bar{X}_i = \frac{1}{nN}\sum_{i=1}^{N}\sum_{j=1}^{n}X_{ij},$$

$$\widehat{\mathbf{E}[s^2(\theta_i)]} = \frac{1}{N}\sum_{i=1}^{N}S_i^2 = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1}{n-1}\sum_{j=1}^{n}\left(X_{ij}-\bar{X}_i\right)^2\right].$$

To estimate $\mathbf{Var}[m(\theta_i)]$, we need to use the law of total variance (or unconditional variance formula), which states that

$$\mathbf{Var}(\bar{X}_i) = \mathbf{E}[\mathbf{Var}(\bar{X}_i|\theta_i)] + \mathbf{Var}[\mathbf{E}(\bar{X}_i|\theta_i)].$$

Noting that $\mathbf{Var}[\mathbf{E}(\bar{X}_i|\theta_i)] = \mathbf{Var}[m(\theta_i)]$ and $\mathbf{E}[\mathbf{Var}(\bar{X}_i|\theta_i)] = \mathbf{E}[s^2(\theta_i)]/n$ (*try to show these as an exercise!*), we have

$$\mathbf{Var}(\bar{X}_i) = \frac{1}{n}\mathbf{E}[s^2(\theta_i)] + \mathbf{Var}[m(\theta_i)].$$

We can now estimate the quantities in the equation above using their unbiased estimators, i.e.,

$$\frac{1}{N-1}\sum_{i=1}^{N}\left(\bar{X}_i-\bar{X}\right)^2 = \frac{1}{nN}\sum_{i=1}^{N}\left[\frac{1}{n-1}\sum_{j=1}^{n}\left(X_{ij}-\bar{X}_i\right)^2\right] + \widehat{\mathbf{Var}[m(\theta_i)]}$$

which rearranges to

$$\widehat{\mathbf{Var}[m(\theta_i)]} = \frac{1}{N-1}\sum_{i=1}^{N}\left(\bar{X}_i-\bar{X}\right)^2 - \frac{1}{nN}\sum_{i=1}^{N}\left[\frac{1}{n-1}\sum_{j=1}^{n}\left(X_{ij}-\bar{X}_i\right)^2\right].$$

Hence, the credibility estimate for risk i based on the EBCT Model 1 is $\boxed{Z\bar{X}_i + (1-Z)\bar{X}}$, where

$$\boxed{Z = \frac{n}{n + \frac{\widehat{\mathbf{E}[s^2(\theta_i)]}}{\widehat{\mathbf{Var}[m(\theta_i)]}}}}.$$

Example 3 The data reported in the table below give the aggregate claims in six successive years from five industrial fire insurance policies in the portfolio of an insurer. The amounts are adjusted for inflation and the unit is in \$1000.

Policy	Year (j)					
number (i)	1	2	3	4	5	6
1	103	73	32	101	78	87
2	112	138	29	93	103	71
3	135	155	121	123	77	139
4	91	105	108	111	116	81
5	67	132	65	93	118	89

Using EBCT Model 1, calculate the credibility premium for the following year in each policy.

Solution: First, calculate the sample mean and sample variance of the claim amounts across $n = 6$ years for each policy $i = 1, 2, \cdots, N(= 5)$.

Policy	Year (j)						$\bar{X}_i$	S_i^2
number (i)	1	2	3	4	5	6	$= \frac{1}{n}\sum_{j=1}^{n} X_{ij}$	$= \frac{1}{n-1}\sum_{j=1}^{n}(X_{ij} - \bar{X}_i)^2$
1	103	73	32	101	78	87	79	674
2	112	138	29	93	103	71	91	1408.4
3	135	155	121	123	77	139	125	704
4	91	105	108	111	116	81	102	176.8
5	67	132	65	93	118	89	94	723.2

We estimate the following quantities:

$$\widehat{\mathbf{E}[m(\theta_i)]} = \frac{79 + 91 + 125 + 102 + 94}{5} = 98.2,$$

$$\widehat{\mathbf{E}[s^2(\theta_i)]} = \frac{674 + 1408.4 + 704 + 176.8 + 723.2}{5} = 737.28$$

$$\widehat{\mathbf{Var}[m(\theta_i)]} = \frac{(79-98.2)^2+\cdots+(94-98.2)^2}{4} - \frac{737.28}{6} = 169.82$$

Hence, the credibility factor is estimated as

$$Z = \frac{6}{6+\frac{737.28}{169.82}} = \frac{8491}{14635} = 0.5802.$$

The credibility premium for each policy is calculated as follows:

Policy number	Credibility Premium
1	$0.5802 \times 79 + 0.4198 \times 98.2 = 87.06$
2	$0.5802 \times 91 + 0.4198 \times 98.2 = 94.02$
3	$0.5802 \times 125 + 0.4198 \times 98.2 = 113.75$
4	$0.5802 \times 102 + 0.4198 \times 98.2 = 100.40$
5	$0.5802 \times 94 + 0.4198 \times 98.2 = 95.76$